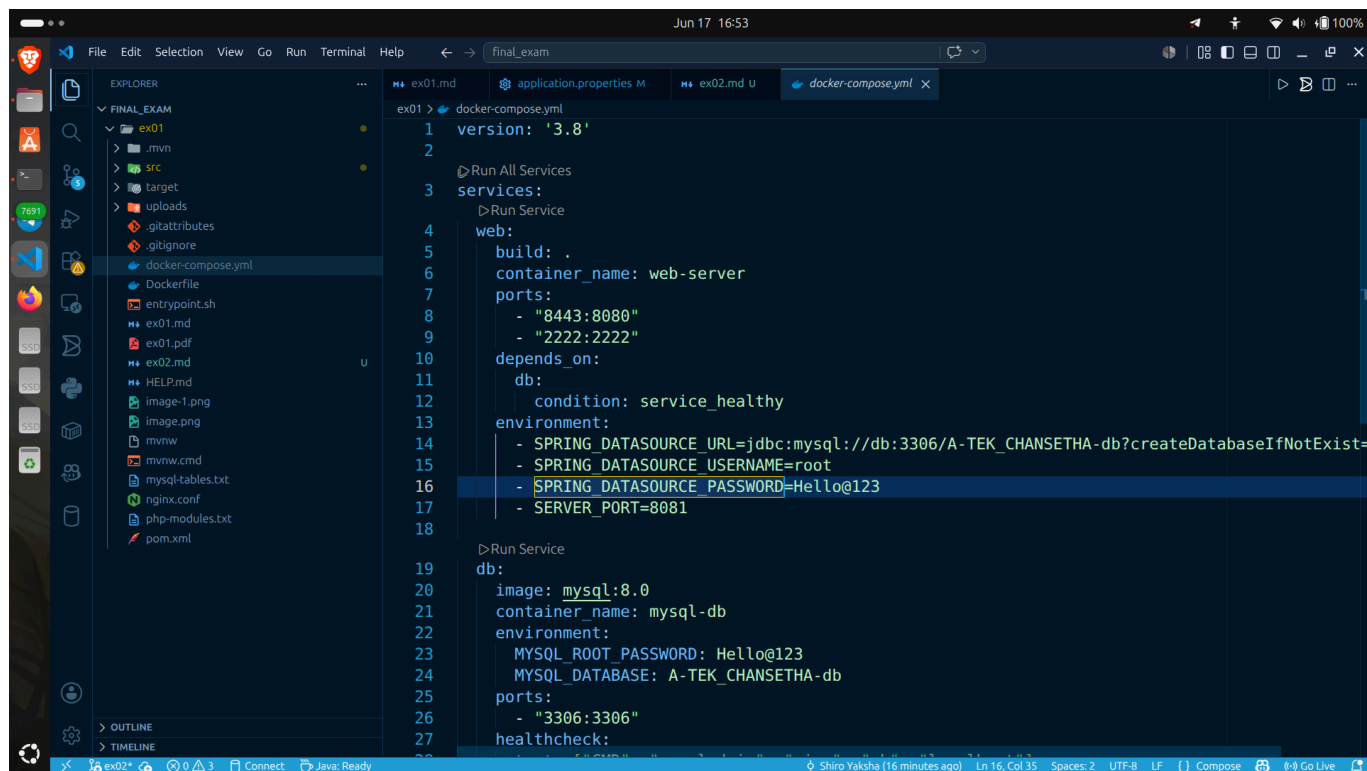


## Screenshots of the work

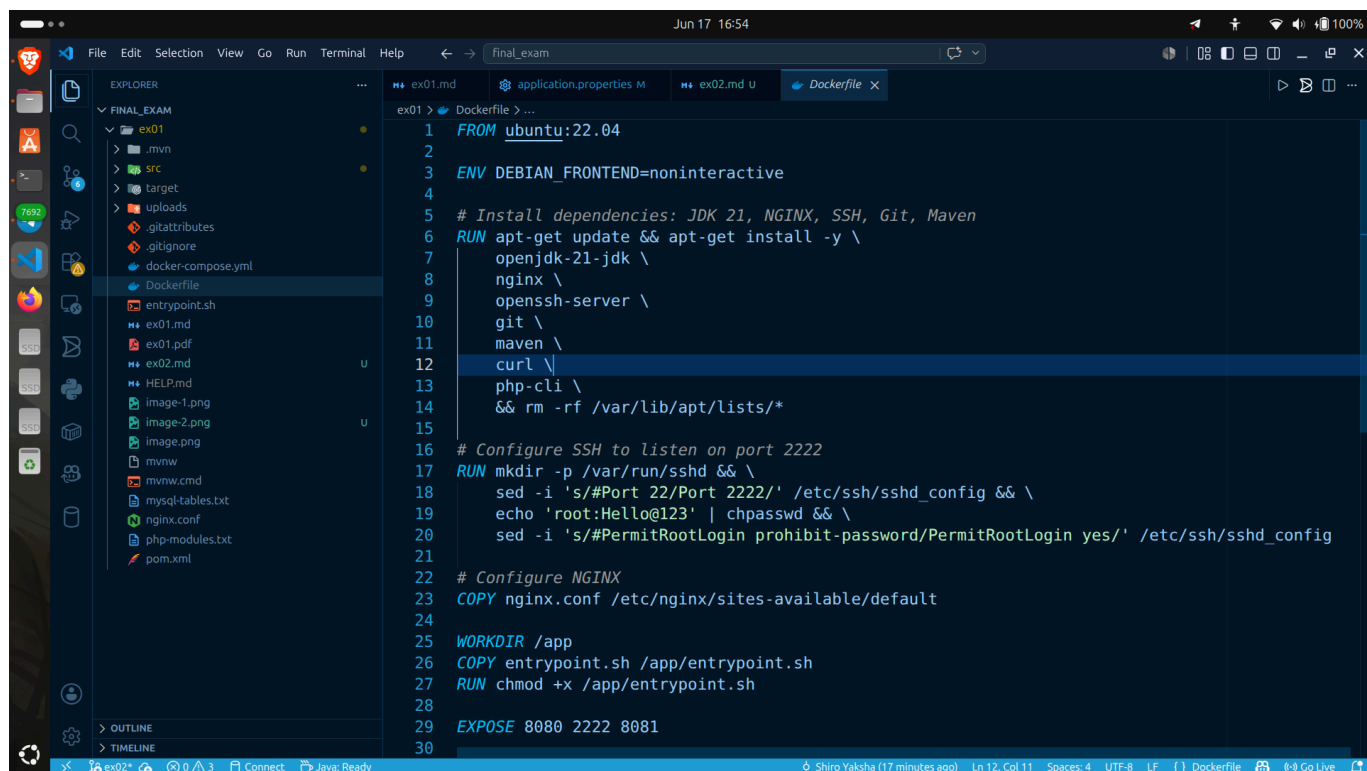
## docker-compose.yml



The screenshot shows the VS Code editor with the 'final\_exam' project open. The 'EXPLORER' sidebar on the left shows the project structure, including a 'docker-compose.yml' file. The main editor area displays the content of 'docker-compose.yml'.

```
1 version: '3.8'
2
3 >Run All Services
4 services:
5   >Run Service
6   web:
7     build: .
8     container_name: web-server
9     ports:
10      - "8443:8080"
11      - "2222:2222"
12     depends_on:
13      db:
14        condition: service_healthy
15     environment:
16      - SPRING_DATASOURCE_URL=jdbc:mysql://db:3306/A-TEK_CHANSETHA-db?createDatabaseIfNotExist=
17      - SPRING_DATASOURCE_USERNAME=root
18      - SPRING_DATASOURCE_PASSWORD=Hello@123
19      - SERVER_PORT=8081
20
21   >Run Service
22   db:
23     image: mysql:8.0
24     container_name: mysql-db
25     environment:
26      MYSQL_ROOT_PASSWORD: Hello@123
27      MYSQL_DATABASE: A-TEK_CHANSETHA-db
28     ports:
29      - "3306:3306"
30     healthcheck:
```

## DockerFile



The screenshot shows the VS Code editor with the 'final\_exam' project open. The 'EXPLORER' sidebar on the left shows the project structure, including a 'Dockerfile' file. The main editor area displays the content of 'Dockerfile'.

```
1 FROM ubuntu:22.04
2
3 ENV DEBIAN_FRONTEND=noninteractive
4
5 # Install dependencies: JDK 21, NGINX, SSH, Git, Maven
6 RUN apt-get update && apt-get install -y \
7     openjdk-21-jdk \
8     nginx \
9     openssh-server \
10    git \
11    maven \
12    curl \
13    php-cli \
14    && rm -rf /var/lib/apt/lists/*
15
16 # Configure SSH to listen on port 2222
17 RUN mkdir -p /var/run/sshd && \
18     sed -i 's/#Port 22/Port 2222/' /etc/ssh/sshd_config && \
19     echo 'root:Hello@123' | chpasswd && \
20     sed -i 's/#PermitRootLogin prohibit-password/PermitRootLogin yes/' /etc/ssh/sshd_config
21
22 # Configure NGINX
23 COPY nginx.conf /etc/nginx/sites-available/default
24
25 WORKDIR /app
26 COPY entrypoint.sh /app/entrypoint.sh
27 RUN chmod +x /app/entrypoint.sh
28
29 EXPOSE 8080 2222 8081
30
```

## nginx.conf

The screenshot shows the VS Code editor with the 'final\_exam' project open. The Explorer panel on the left shows the project structure, including files like ex01.md, ex02.md, and nginx.conf. The main editor window displays the contents of 'nginx.conf'.

```

1 server {
2     listen 8080 default_server;
3     listen [::]:8080 default_server;
4
5     server_name _;
6
7     location / {
8         proxy_pass http://127.0.0.1:8081;
9         proxy_set_header Host $host;
10        proxy_set_header X-Real-IP $remote_addr;
11        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
12        proxy_set_header X-Forwarded-Proto $scheme;
13    }
14 }
15

```

## entrypoint.sh

The screenshot shows the VS Code editor with the 'final\_exam' project open. The Explorer panel on the left shows the project structure, including files like ex01.md, ex02.md, and entrypoint.sh. The main editor window displays the contents of 'entrypoint.sh'.

```

1 #!/bin/bash
2 # Start SSH daemon
3 /usr/sbin/sshd
4
5 # Start NGINX
6 service nginx start
7
8 # Clone the Spring Boot project
9 cd /app
10 if [ ! -d "DevOps-Final-Exam" ]; then
11     echo "Cloning Spring Boot project..."
12     git clone -b ex01 https://github.com/ShiroYaksha01/DevOps-Final-Exam.git
13 fi
14 cd DevOps-Final-Exam
15
16 # Build the project
17 echo "Building the project..."
18 mvn clean package -DskipTests
19
20 # Run the project
21 echo "Starting Spring Boot..."
22 java -jar target/*.jar
23
24

```

## Result outputs

### docker ps

```

shiro@shiroyaksha:~/Documents/DevOps/final_exam/ex01$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS
e852e01ce712   ex01-web      "/app/entrypoint.sh"    5 minutes ago Up 5 minutes  0.0.0.0:2222->2222/tcp, [::]:2222->2222/tcp, 8081/tcp, 0.0.0.0:8443->8080/tcp, [::]:8443->8080/tcp
044d11d9a68c   mysql:8.0     "docker-entrypoint.s..." 18 minutes ago Up 18 minutes (healthy) 0.0.0.0:3306->3306/tcp, [::]:3306->3306/tcp, 33060/tcp

```

## curl -I http://localhost:8443

```
shiro@shiroyaksha:~/Documents/DevOps/final_exam/ex01$ curl -I http://localhost:8443
HTTP/1.1 200
Server: nginx/1.18.0 (Ubuntu)
Date: Wed, 17 Jun 2026 09:41:18 GMT
Content-Type: text/html; charset=UTF-8
Connection: keep-alive
Content-Language: en-US
```

## docker exec mysql-db mysql -uroot -pHello@123 -e "SHOW DATABASES;"

```
shiro@shiroyaksha:~/Documents/DevOps/final_exam/ex01$ docker exec mysql-db mysql -uroot -pHello@123 -e "SHOW DATABASES;"
mysql: [Warning] Using a password on the command line interface can be insecure.
Database
A-TEK_CHANSETHA-db
information_schema
mysql
performance_schema
sys
```

## ssh root@localhost -p 2222,

```
shiro@shiroyaksha:~/Documents/DevOps/final_exam/ex01$ ssh root@localhost -p 2222
The authenticity of host '[localhost]:2222 ([127.0.0.1]:2222)' can't be established.
ED25519 key fingerprint is SHA256:3/YcJNHZ41Y9b5crQBzIB8kwaywKSxI8IRvEIjxJi8c.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[localhost]:2222' (ED25519) to the list of known hosts.
root@localhost's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.17.0-35-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

root@e852e01ce712:~#
```

## mysql table

```
shiro@shiroyaksha:~/Documents/DevOps/final_exam/ex01$ cat mysql-tables.txt
Tables_in_A-TEK_CHANSETHA-db
profiles
templates
```

## php-module

```
shiro@shiroyaksha:~/Documents/DevOps/final_exam/ex01$ cat php-modules.txt | head -n 15
[PHP Modules]
calendar
Core
ctype
date
exif
FFI
fileinfo
filter
ftp
gettext
hash
iconv
json
libxml
```